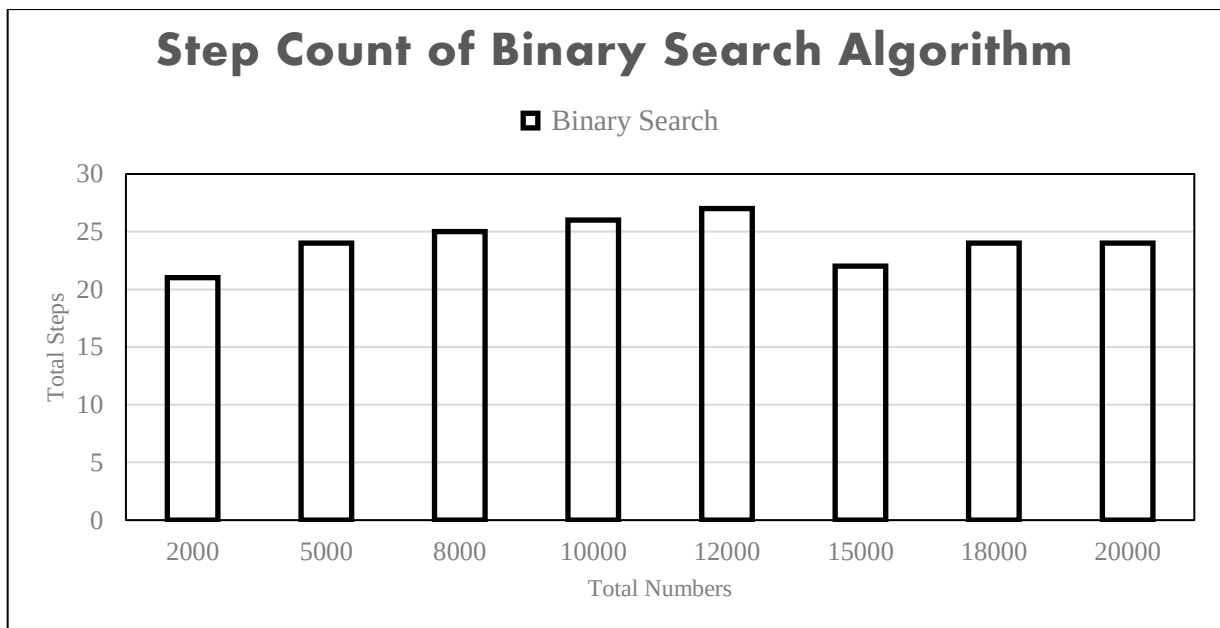
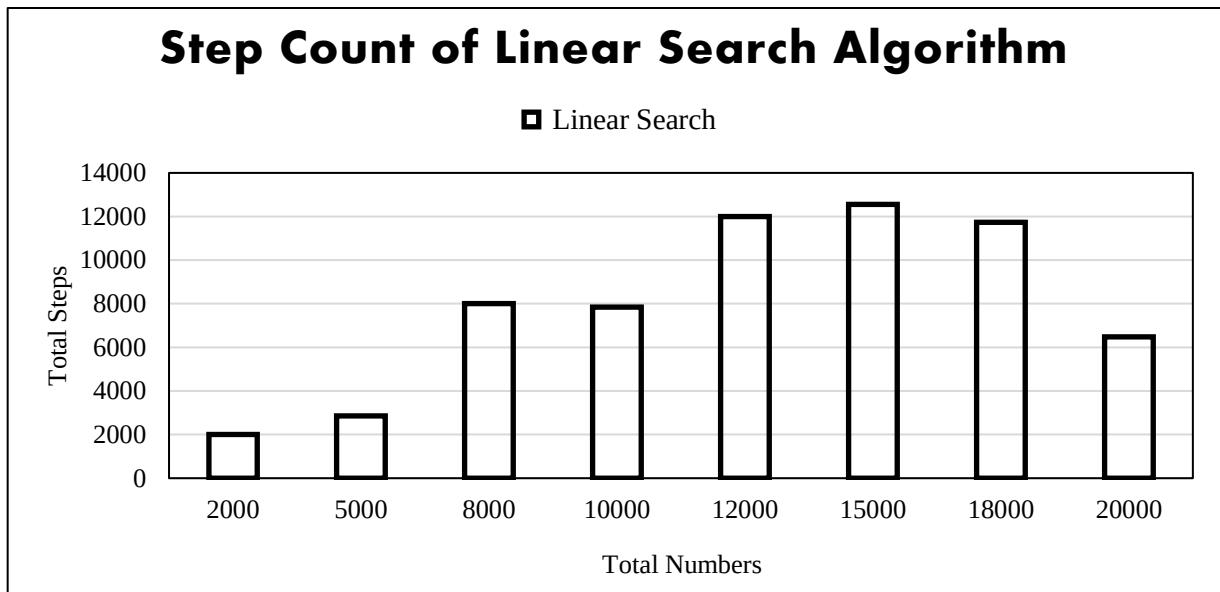


Output: (Steps)

Pos=Position

Num Name	2000		5000		8000		10000		12000		15000		18000		20000	
-	Pos	Step	Pos	Step	Pos	Step	Pos	Step	Pos	Step	Pos	Step	Pos	Step	Pos	Step
Linear	-1	2001	2849	2849	-1	8001	7842	7843	-1	12001	12559	12560	11736	11737	6478	6479
Binary		21		24		25			26			27				22

- Graphical Interpretation:



❖ Implemented Code (C++):

```
#include<bits/stdc++.h>
using namespace std;

ll step=0;
int Linear_Search(vector<int>A,int find,int i)
{
    int n=A.size();
    if(i==n)
    {
        step++;
        return -1;
    }
    if(A[i]==find)
    {
        step++;
        return i;
    }
    step++;
    return Linear_Search(A,find,i+1);
}

int Binary_Search(vector<int>A,int find,int low,int high)
{
    if(low==high)
    {
        step++;
        if(A[low]==find)
        {
            step++;
            return low;
        }
        return -1;
    }
    step++;
    int mid=(low+high)/2;
    if(A[mid]==find)
    {
        step++;
        return mid;
    }
    else if(A[mid]<find)
    {

```

```

        step++;
        return Binary_Search(A,find,mid+1,high);
    }
    else
    {
        step++;
        return Binary_Search(A,find,low,mid-1);
    }
}
int main()
{
    srand(time(0));
    vector<int>A[8],B={2000,5000,8000,10000,12000,15000,18000,20000};
    for(int i=0;i<8;i++)
    {
        ifstream ip;
        ip.open("input1.txt");
        for(int j=0;j<B[i];j++)
        {
            int x;
            ip>>x;
            A[i].push_back(x);
        }
        ip.close();
    }
    ofstream op;
    op.open("output.txt");
    for(int i=0;i<8;i++)
    {
        op<<"Total value: ";
        op<<B[i]<<"\n";
        int a=rand();
        op<<"Linear Search: \n";
        op<<"Position: "<<Linear_Search(A[i],a,0)<<" Total Steps: "<<step<<"\n";
        step=0;
        op<<"Binary Search: \n";
        op<<"Position: "<<Binary_Search(A[i],a,0,A[i].size())<<" Total Steps: "<<step<<"\n\n\n";
        step=0;
    }
}

```